FINANCIAL PERFORMANCE ANALYSIS REPORT

MyUSA Bank | 2022 2023

Profitability, efficiency, returns and stock performance

Python | pandas | SQLite | SQL | Matplotlib

Prepared by Abdur Raffay

31 August 2026

Portfolio analysis based on a simulated financial dataset.

Executive Summary

The analysis shows a mixed 2023 performance picture. MyUSA Bank improved its net interest margin, but this did not translate into stronger bottom line profitability. Average net income, ROA and ROE all declined, while the cost to income ratio increased slightly. Average stock price was almost unchanged and showed no meaningful linear relationship with the financial performance measures tested.

$1.562m	3.14%	6.38%	28.28%
Average net income	**Net interest margin**	**Return on equity**	**Cost to income**
2023 | 3.29% YoY	2023 | +2.98% YoY	2023 | 3.48% YoY	2023 | +1.29% YoY

Key Findings

- Average net income decreased from $1.616m in 2022 to $1.562m in 2023, a 3.29% decline.
- Net interest margin improved from 3.05% to 3.14%, a 2.98% increase, indicating stronger interest spread performance despite weaker overall profitability.
- ROA fell from 1.80% to 1.74% and ROE fell from 6.61% to 6.38%, showing weaker returns from both assets and shareholder equity.
- The cost to income ratio increased from 27.91% to 28.28%. Because a higher ratio indicates more income being absorbed by operating costs, this represents a slight deterioration in efficiency.
- Average stock price was effectively flat at $126.48 in 2022 and $126.35 in 2023. Pearson correlations between stock price and the tested performance metrics were all close to zero.
- No negative net income periods were present in the cleaned dataset.

Annual Performance Comparison

Metric	2022	2023	Change
Net income	$1.616m	$1.562m	3.29%
Net interest margin	3.05%	3.14%	+2.98%
ROA	1.80%	1.74%	3.25%
ROE	6.61%	6.38%	3.48%
Cost to income	27.91%	28.28%	+1.29%
Stock price	$126.48	$126.35	0.10%

1. Data and Methodology

The project uses a simulated time series dataset for a fictional bank. The cleaned dataset contains 499 dated observations: 250 in 2022 and 249 in 2023. The SQLite database contains the same 499 unique date records, with no duplicate dates detected.

Data Quality

- The cleaned dataset contains 49 missing values in interest income and in the financial metrics derived from it, including net income, net interest income, NIM, ROA and ROE.
- Missing observations were not imputed. pandas and SQL averages therefore use the available non null observations for each metric.
- Net income is available for 226 observations in 2022 and 224 observations in 2023.
- Outliers were retained for analysis rather than automatically removed, because unusual financial periods may be analytically meaningful.
- No duplicate dates are present in the final SQLite table.

Calculated Metrics

Metric	Calculation
Net interest income	Interest income Interest expense
Net interest margin	Net interest income / Average earning assets x 100
Return on assets	Net income / Total assets x 100
Return on equity	Net income / Shareholder equity x 100
Cost to income ratio	Operating expenses / Operating income x 100

Analytical Workflow

The workflow combines relational and statistical analysis. Python and pandas are used for cleaning, validation and higher level analysis; SQLite stores the processed data; SQL identifies key periods and performs aggregation; pandas calculates yearly comparisons and Pearson correlations; and Matplotlib converts the time series into monthly trend visualisations.

Currency Presentation

The supplied dataset does not contain an explicit currency field. For presentation, monetary values are shown in US dollars to remain consistent with the MyUSA Bank context and the project visualisations. This is a reporting convention rather than a verified property of the source data.

2. Profitability and Interest Margin

Net Income

Average net income declined by 3.29% year on year, from $1.616m in 2022 to $1.562m in 2023. The monthly series shows a weaker middle section of 2023, with the lowest monthly average occurring in September 2023 at approximately $1.406m. The strongest monthly average occurred in July 2022 at approximately $1.752m.

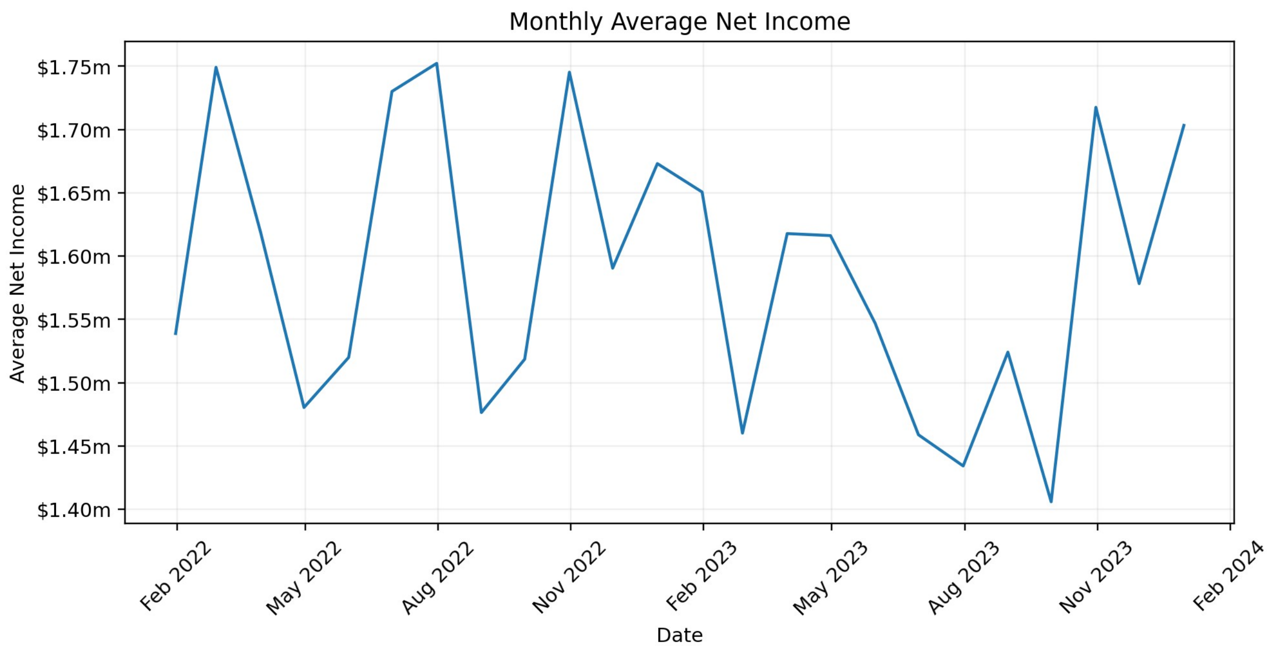

Figure 1. Monthly average net income. Monthly aggregation reduces day to day volatility while preserving the overall direction of profitability.

The SQL analysis also identifies a single period maximum net income of $3.999m on 11 October 2023. This illustrates why individual high observations and monthly averages should be interpreted separately: a strong single period did not prevent the 2023 annual average from finishing below 2022.

Net Interest Margin

Net interest margin moved in the opposite direction to net income. The annual average increased from 3.05% in 2022 to 3.14% in 2023, a 2.98% improvement. Monthly NIM peaked at 3.42% in July 2023 and reached its lowest monthly average at 2.83% in June 2022. The highest individual NIM observation was 4.75% on 4 May 2023.

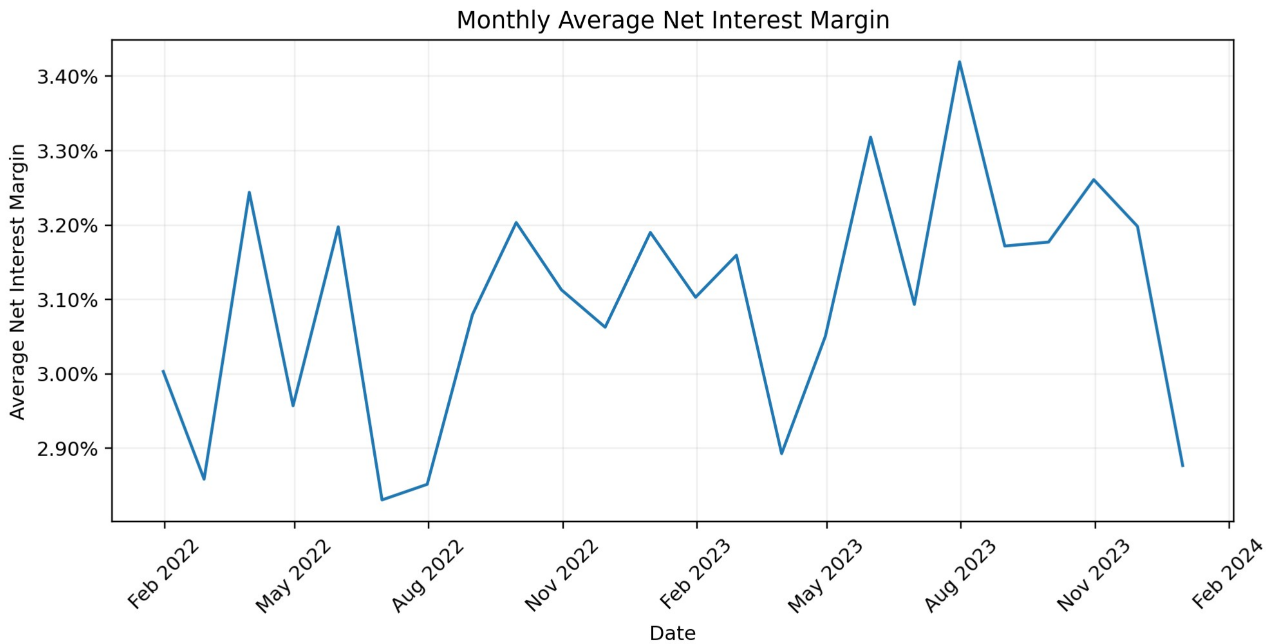

Figure 2. Monthly average net interest margin.

This divergence is one of the report's main findings. The bank generated a stronger interest margin in 2023, but overall profitability still weakened. The dataset alone does not establish why; changes in other income, expenses, asset levels or the simulated data generating process could contribute, so no causal explanation is assumed.

3. Returns and Operating Efficiency

ROA and ROE

Returns weakened alongside net income. Average ROA declined from 1.80% in 2022 to 1.74% in 2023, while average ROE declined from 6.61% to 6.38%. The year on year reductions were 3.26% and 3.48% respectively.

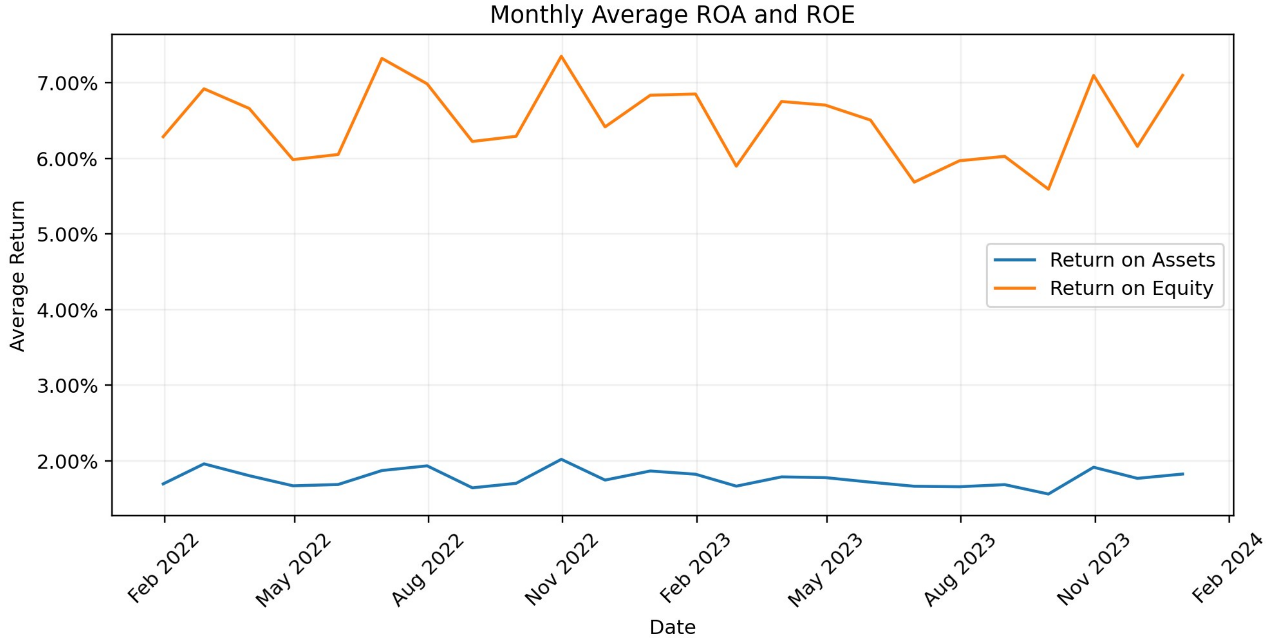

Figure 3. Monthly average return on assets and return on equity.

Both monthly return measures reached their highest averages in October 2022: approximately 2.02% for ROA and 7.35% for ROE. Both reached their lowest monthly averages in September 2023: approximately 1.56% and 5.59%. The shared September trough aligns with the weakest monthly net income period and supports the conclusion that the profitability pressure was visible across multiple return measures.

Cost to Income Ratio

The annual cost to income ratio increased from 27.91% to 28.28%, a 1.29% rise. Because lower values generally represent stronger operating efficiency, the annual movement indicates a modest deterioration in efficiency during 2023.

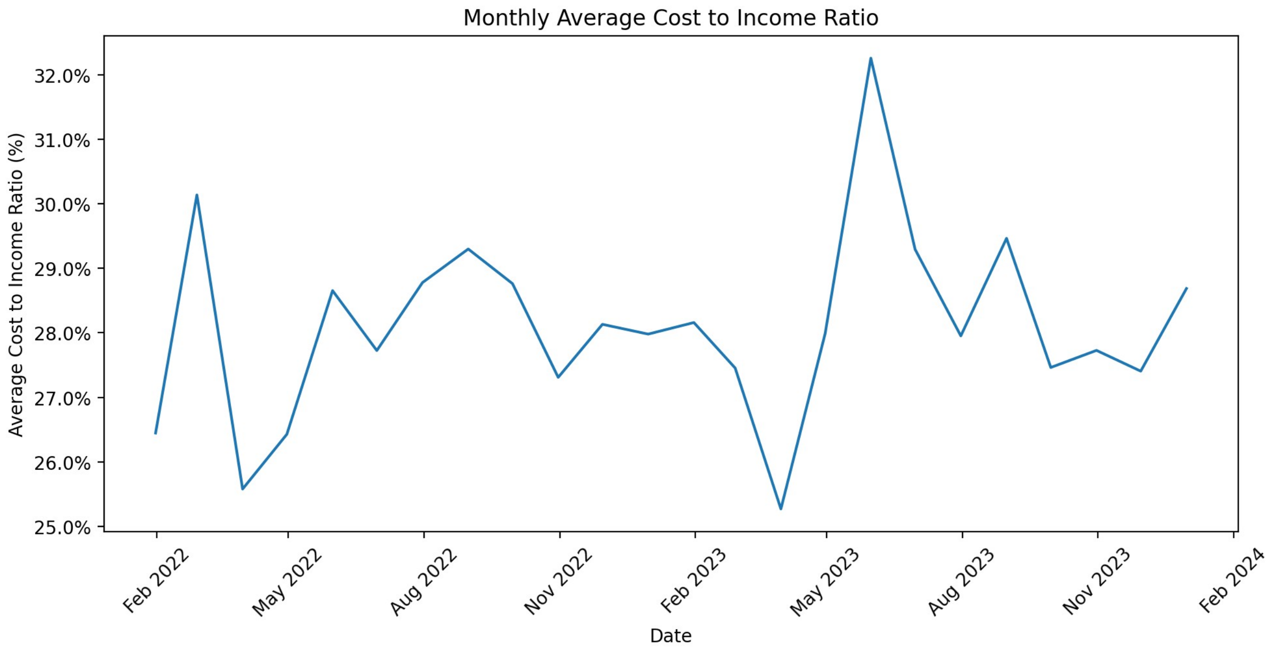

Figure 4. Monthly average cost to income ratio. Lower values are interpreted as more efficient.

Monthly efficiency was volatile. The best monthly average was 25.27% in March 2023, while the worst was 32.25% in May 2023. SQL identified the lowest individual cost to income ratio as 14.22% on 17 October 2023. The sharp monthly movement around spring 2023 is therefore more informative than the small annual change alone.

4. Stock Price and Financial Performance

Average stock price was almost unchanged between the two years, moving from $126.48 in 2022 to $126.35 in 2023, a decline of only 0.10%. Monthly average stock price peaked at approximately $146.05 in December 2022 and reached its lowest monthly average at approximately $108.32 in February 2023.

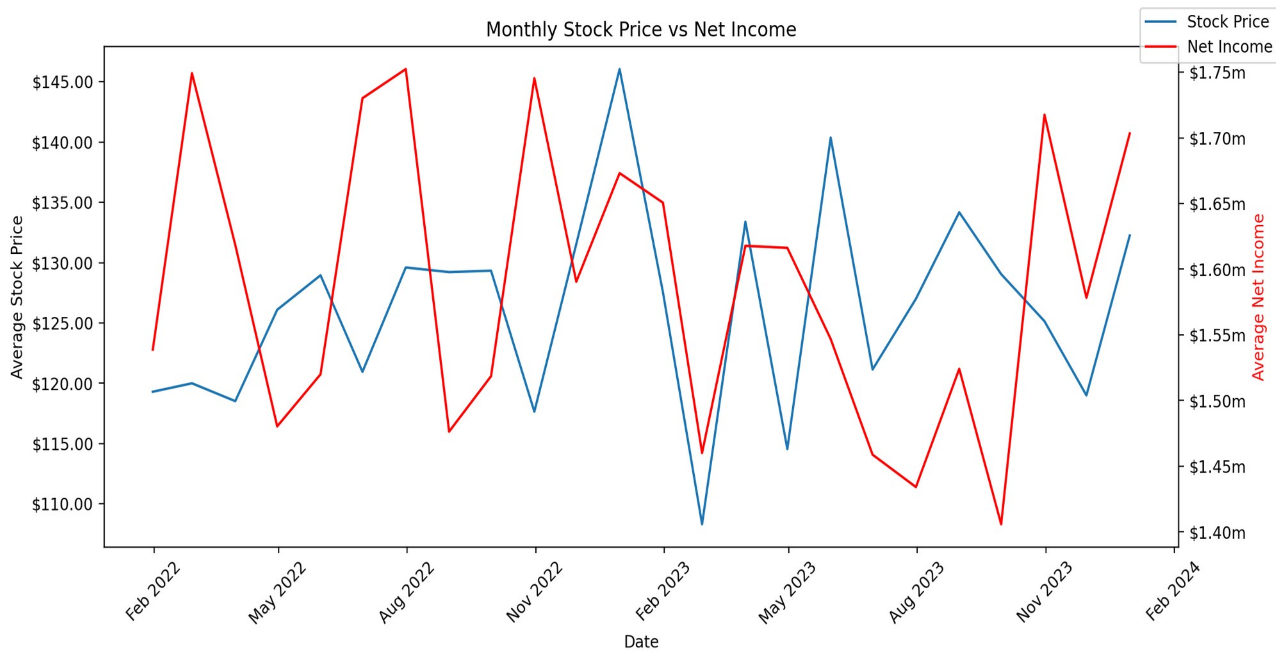

Figure 5. Monthly average stock price compared with monthly average net income. The two axes use different units and are shown for directional comparison only.

Pearson Correlation Results

Metric vs Stock Price	Pearson r	Interpretation
Net income	+0.047	Negligible
Net interest margin	0.011	Negligible
ROA	+0.040	Negligible
ROE	+0.077	Negligible
Cost to income	0.029	Negligible

All correlations are extremely close to zero. ROE has the largest absolute coefficient at only +0.077, followed by net income at +0.047. Within this simulated two year dataset, there is therefore no meaningful linear relationship between stock price and the tested financial metrics. This does not rule out non linear relationships or lagged effects, and correlation does not establish causation.

5. Selected SQL Findings

The SQL stage complements the annual and monthly analysis by identifying individual periods with exceptional values. These records are useful for investigation but should not be treated as representative of the broader trend.

Highest Individual Net Income

Date	Net Income
11 Oct 2023	$3.999m
23 Feb 2022	$3.972m
01 Jun 2022	$3.945m
29 Jul 2022	$3.943m
23 Jun 2022	$3.891m

Highest Individual Net Interest Margin

Date	NIM
04 May 2023	4.75%
30 Jun 2022	4.67%
27 Sep 2022	4.64%
04 Oct 2022	4.62%
25 Oct 2022	4.60%

Most Efficient Individual Periods

Date	Cost to Income
17 Oct 2023	14.22%
21 Dec 2023	14.45%
21 Mar 2022	14.63%
21 Jul 2023	14.63%
07 Sep 2022	15.11%

No negative net income observations were found in the cleaned database. This means the analysis concerns changes in the level of profitability rather than a transition between profit and loss.

6. Limitations and Conclusion

Limitations

- The dataset is simulated and does not represent the performance of a real bank.
- The source data does not explicitly state a currency; US dollar formatting is used only as a presentation convention.
- ROA uses total assets at each observation rather than average assets across a reporting period.
- ROE uses shareholder equity at each observation rather than average equity across a reporting period.
- Forty nine observations are missing several income related fields. These values were not imputed, so averages use available observations.
- Monthly averaging improves readability but can hide individual spikes and extreme values.
- The dataset covers only two years, limiting longer term inference.
- Pearson correlation measures linear association only and cannot demonstrate causation.

Conclusion

The overall assessment is that 2023 was a mixed year for the simulated bank. Net interest margin improved, suggesting stronger spread performance, but this improvement was not matched by the bank's broader profitability measures. Average net income, ROA and ROE all declined, while the cost to income ratio increased slightly. The most pronounced weakness appeared around the middle to later part of 2023, with September recording the lowest monthly averages for net income, ROA and ROE.

The stock price remained almost unchanged on an annual average basis and exhibited negligible Pearson correlations with the financial metrics analysed. In this dataset, stock price therefore provides little evidence of moving directly with short term changes in profitability, returns or efficiency.

The project demonstrates an end to end analytical workflow: raw financial data is cleaned and validated, stored in SQLite, queried with SQL, analysed with pandas and communicated through Matplotlib visualisations. The key analytical insight is not that every metric moved in the same direction, but that the bank showed stronger net interest margin alongside weaker overall returns and slightly poorer cost efficiency a mixed performance profile that becomes clear only when multiple measures are considered together.

END OF REPORT